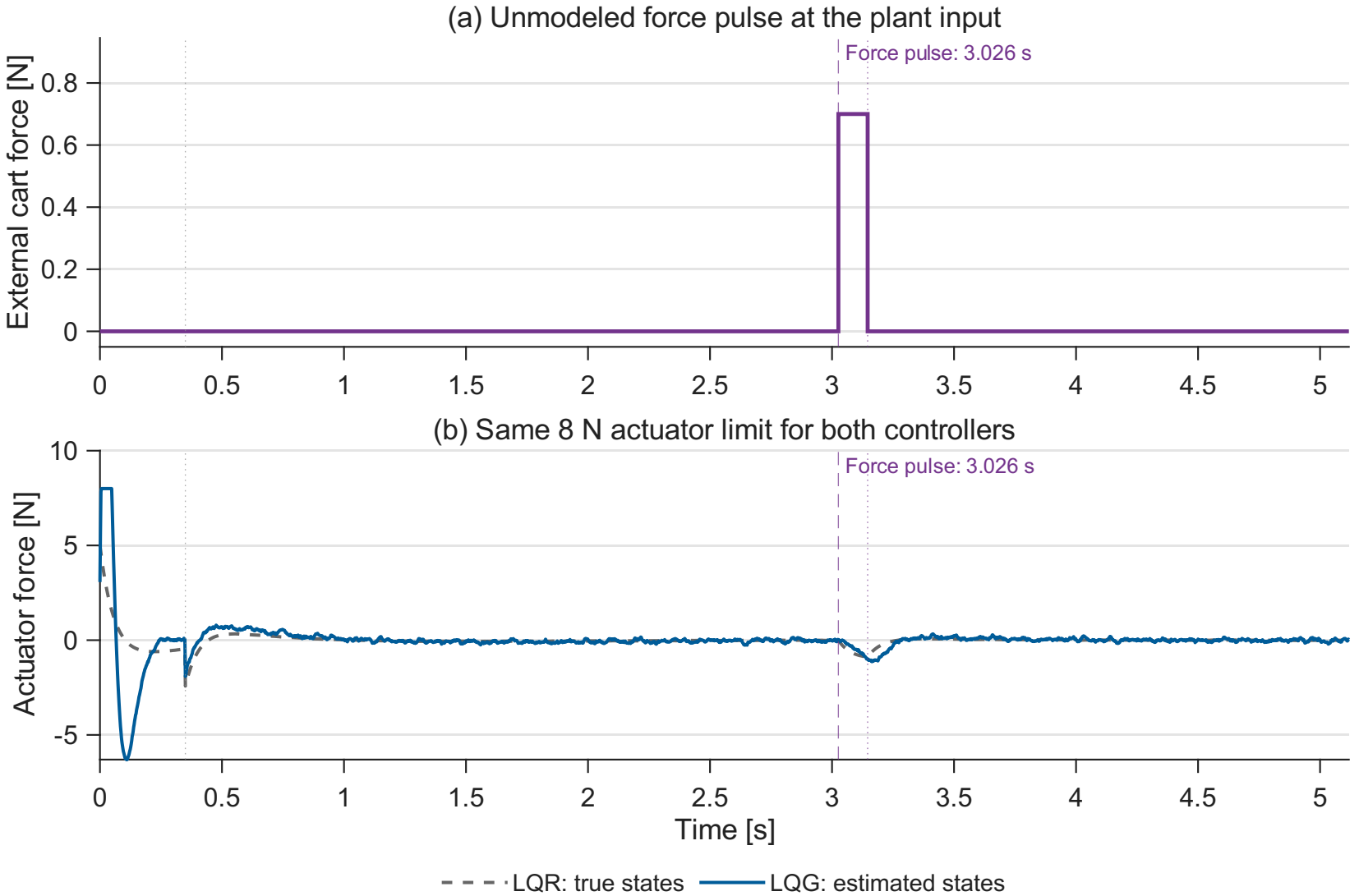

(a) Unmodeled force pulse at the plant input
Force pulse: 3.026 s
External cart force [N]
0.8
0.6
0.4
0.2
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
(b) Same 8 N actuator limit for both controllers
Force pulse: 3.026 s
Actuator force [N]
10
5
0
-5
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Time [s]
LQR: true states
LQG: estimated states